
When a recovery curve fails to validate a decision rule: a single-cell calibration case study

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We re-evaluate a single-cell reporting rule whose 50-cell cutpoint was declared
2 validated from point-estimate recovery. The original memo and calibration criteria
3 are reproduced in the paper. Direct evaluation on two colorectal single-cell cohorts
4 yields no qualifying cutpoint under pooled-tissue draws; reference-only candidates
5 change with cohort, count grid and aggregation. At 50 cells, discrimination is
6 75.5–81.0% in one cohort and 45.0–55.0% in the other, against an 80% requirement.
7 Controlled reuse of the same simulation draws separates grid effects from random-
8 stream changes. The interval target is a draw-dependent benchmark, so these
9 results test the recorded calibration specification rather than population coverage.
10 A targeted diagnostic finds 37.5–67.0% null rejection at five cells, explaining why
11 high low-count exclusion rates cannot be read as reliable power. Biological controls
12 also fail to establish the intended mechanistic separation. We explain the recovery
13 curve using an elementary estimator identity and use established trial-estimator
14 relationships as implementation checks. The contribution is a reproducible failed-
15 calibration case study: threshold validation requires direct evaluation of intervals,
16 null rejection and abstention, with the target and sampling unit declared in advance.

17 1 Motivation: recovery and decision-rule validation

18 This paper revisits a specific validation claim: a single-cell analysis reported that an estimator
19 recovered an imposed effect and treated this as validation of a 50-cell reporting cutpoint. The original
20 design required interval inclusion and discrimination, which the recovery ratio did not measure.
21 Section 2 reproduces the claim and criteria, then tests them directly. The empirical contribution is
22 the failed calibration and its sensitivity to sampling and aggregation. The algebra used to explain
23 the original comparison is elementary; the trial examples are worked checks of established estimator
24 relationships.

25 Our motivating application separates two explanations for differentiation-marker loss in colorectal
26 cancer: fewer marker-producing cells (a *compositional* change), or lower output from the cells that
27 remain (an *intrinsic* change). These explanations suggest different biological interpretations. An
28 empirical comparison of per-cell means requires the relevant population to be observed in both tissues.
29 Let f_N, f_T denote the relevant cell fractions in reference and diseased tissue, and m_N, m_T their
30 mean per-cell output. A reference-based decomposition, related to demographic standardisation and
31 Oaxaca–Blinder decompositions [Kitagawa, 1955, Oaxaca, 1973, Blinder, 1973], is

$$\underbrace{(f_T - f_N)m_N}_{\text{compositional}} + \underbrace{f_N(m_T - m_N)}_{\text{intrinsic}} + \underbrace{(f_T - f_N)(m_T - m_N)}_{\text{interaction}} = f_T m_T - f_N m_N. \quad (1)$$

32 The interaction is reported separately. If no relevant cells are observed in an arm, its sample mean is
33 undefined; the identity cannot supply the missing mean. In a synthetic population with no such cells,

the corresponding conditional mean itself has no referent. Substituting zero in either case turns an unavailable contrast into an apparent effect.

The implemented reporting rule returns an estimate at $n \geq 50$, flags a wide interval at $20 \leq n < 50$, and returns NOT ESTIMABLE below 20, where n is the relevant-cell count in diseased tissue. Unavailable reported contrasts are stored as None, not zero. The diagnostic oracle calculations examined below also retain raw arithmetic before this reporting gate. The question is whether the earlier recovery evidence established the interval-performance requirements used to justify the cutpoints.

Relation to existing validation methods. Simulation-study methodology already separates estimands, methods and performance measures [Burton et al., 2006, Morris et al., 2019]. Generator–analysis alignment is discussed in congeniality [Meng, 1994], plasmode and semi-synthetic benchmarks [Franklin et al., 2014, Curth et al., 2021], interventional evaluation [Gentzel et al., 2019] and the inverse crime [Wirgin, 2004, Kaipio and Somersalo, 2007]; generators can also disadvantage good estimators [Shaw et al., 2026]. Our analysis applies these established distinctions to a documented reporting-rule decision. For clinical trials, external-control validation already examines bias and type-I error using held-out studies [Ventz et al., 2019]. Prognostic adjustment with historical data retains trial randomisation and evaluates precision gains [Schuler et al., 2022]; target-trial emulation states the causal question and design explicitly [Hernán and Robins, 2016]. These are different uses of a patient model and need different validation experiments. Our case contributes an inspectable failure analysis relevant to that design problem, not evidence of failure in those methods. Simulation-based calibration evaluates posterior ranks [Talts et al., 2018, Modrák et al., 2025]. Appendix D provides further context.

2 Direct evaluation does not justify the abstention rule

The claim being re-evaluated. The project gate memo, Section 15, states: “The pre-committed cutpoint is validated on real cells.” Its evidence is recovery between 0.85 and 1.08 above the 50-cell reporting boundary and 1.18 in the 20–49-cell band, based on six draws per setting at 1,200 cells per arm. That comparison is an observed/target point-estimate ratio. In contrast, the design specification, Section 4, requires at least 90% interval inclusion and 80% exclusion of zero at a halving of expression. These passages document the mismatch within this project; they do not establish a general practice. Appendix D provides the source locations and preserves their status as an internal claim.

Target, interval and operating requirements. The design document fixed a detectable halving of per-cell output ($s = 0.5$), interval inclusion of the stored reference in at least 90% of defined outputs, and discrimination of at least 80%, defined as the probability that the interval excludes zero at that effect. We call the first rate benchmark inclusion (“coverage” in the saved tables). The candidate *ok* is the first evaluated count or bin meeting both criteria; *wide* is the first meeting coverage alone. The committed values are 50 and 20. Candidates do not require every larger count to qualify and are not certified thresholds.

The calibration uses paired subsets of the published colorectal atlas [Lee et al., 2020]: ten patients from GSE132465 (SMC) and six from GSE144735 (KUL3). We analyse GUCA2A in the differentiated population under fixed lineage labelling. Each draw samples, with replacement, 2,000 cells per arm from an eligible two-patient pool; the reference mature fraction is $f_N = 0.40$. The implemented pool combines cells from both tissues before drawing either synthetic arm. Because the design document did not specify tissue eligibility, we also evaluate a reference-tissue-only pool. Neither reconstructs the real tissue contrast. A nominal 95% percentile interval resamples mature cells in both arms, with fractions held fixed and 200 inner draws unless stated otherwise. Undefined intervals are excluded from rate denominators and reported as abstentions (Appendix B).

The stored “parametric” intrinsic reference is

$$i_{\text{req}}(D_N; s) = f_N(s - 1)\bar{Y}_N, \quad (2)$$

where $\bar{Y}_N$ is the mature-cell mean of the *drawn* reference arm. Thus the shift is fixed but this benchmark varies across draws; it is not the fixed source-pool parameter $f_N(s - 1)\mu_{\text{pool}}$. Benchmark inclusion below uses Equation 2. This is a legitimate draw-specific performance question, but it

differs from fixed-population coverage. We preserve that specification to test the original claim; a population-coverage study requires a separately fixed target and sampling law.

Controlled comparison of counts, bins and source pools. Historical grids returned reference-only binned candidates of 100, 90 and 70, but changed random streams, bin boundaries and replicate budgets together (Appendix B). To isolate these choices, we simulated the union of counts once per cohort, pool and each of eight seeds, with 200 outer replicates and 200 inner bootstrap draws. Every grid summary reuses the same outputs at shared counts. The committed grid reaches $\{0, 20, 40, 100, 200, 400, 800\}$; filled and dense grids add count support. Fixed bins share boundaries across grids; adaptive bins recalculate them. Binned candidates are median counts within the selected bin, not estimated crossing locations.

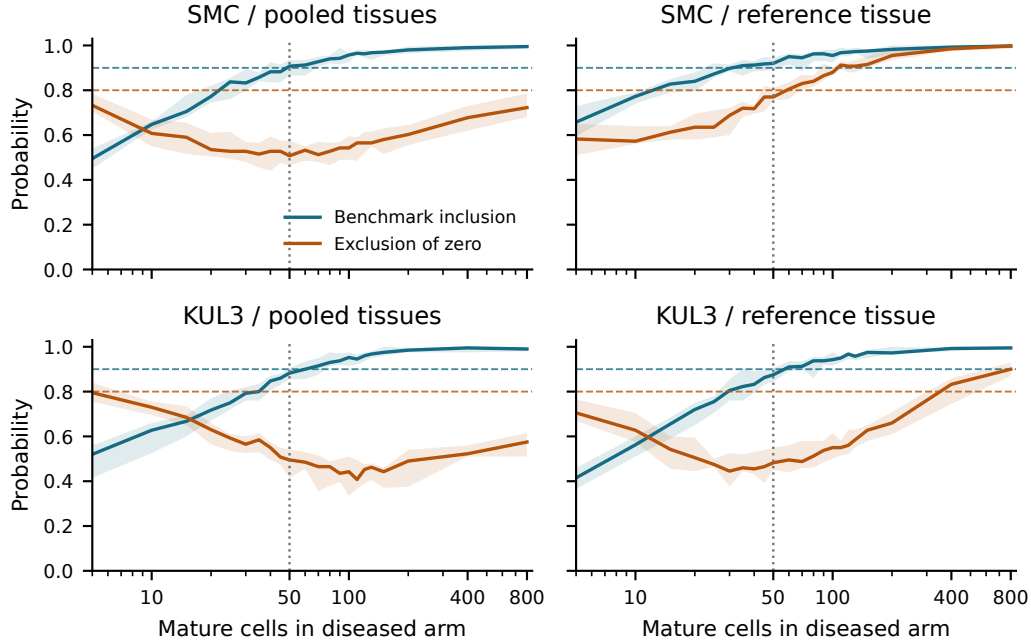


Figure 1: Direct evaluation on the controlled union of counts, without binning. Rows show cohorts; columns show draw pools. Curves are median rates over eight seeds and bands span their minima and maxima, conditional on the observed cohorts. Dashed lines mark the 90% benchmark-inclusion and 80% discrimination targets; the vertical line marks the committed count of 50. Rates condition on defined intervals and coverage uses the draw-dependent reference in Equation 2. Zero-cell settings have no defined interval and are omitted. No pooled-cell candidate qualifies. Per-count reference-only candidates vary by seed and cohort (Table 1).

With common fixed bins, SMC returns 100 on the committed grid and 70 on both expanded grids in all eight seeds. Adaptive bins return 65 on seven filled-grid seeds and 90 on one, while the dense grid returns 70 throughout. KUL3 instead returns 400 on six seeds and 800 on two, with all grid and binning schemes agreeing within each seed. Pooled draws yield no joint candidate in either cohort. The largest binomial Monte Carlo standard error is 0.035, conditional on the fixed cohorts; it does not measure patient-sampling uncertainty.

At the committed count of 50, SMC reference-only discrimination ranges from 75.5% to 81.0% across seeds, with both criteria met on two of eight seeds. KUL3 reference-only discrimination is only 45.0–55.0% at that count. These direct observations make the lack of support for a cohort-independent 50-cell rule more concrete than the selected binned candidates alone.

The per-count summaries in Table 1 supply a useful check on the bin medians. In SMC, the filled grid’s first qualifying actual count is 50 on two seeds, 60 on five and 80 on one; the dense grid adds a candidate of 45 on one seed. Thus a stable binned candidate of 70 is not evidence that 70 is a uniquely determined biological threshold. A bin pools outcomes over several count settings, whereas a per-count rate addresses only that setting. Fixed bins hold the boundaries constant, but adding settings can still change both a bin’s median count and its aggregate performance.

Table 1: Joint coverage-and-discrimination candidates from the controlled union-grid run. Entries give candidates with seed frequencies in parentheses where they vary; otherwise all eight seeds agree. Every grid summary reuses the same draws at shared counts. “None” means no tested count or bin qualifies.

cohort / pool	grid	fixed bins	adaptive bins	per count
SMC / reference	committed	100	100	100
	filled	70	65 (7), 90 (1)	50–80
	dense	70	70	45–80
KUL3 / reference	all three	400 (6), 800 (2)	same	same
Both / pooled	all three	None	None	None

Low-count rejection and nonmonotone qualification. The high exclusion rate at very small counts in Figure 1 is not evidence of reliable low-count power. We performed an exploratory diagnostic at counts 5, 50, 100 and 800: 200 replicates per cohort, pool and shift ($s = 0.5$ and 1), using the first controlled-run seed and the same 200-resample interval. All 16 alternative-setting summaries reproduce their saved counterparts. Under the null ($s = 1$), exclusion of zero is a false positive, and the target is exactly zero under both the draw-dependent and fixed-pool definitions. Table 2 reports both shifts with the same method and valid-pair denominator; all 6,400 diagnostic outputs are finite.

Table 2: Exclusion of zero (%) with the same percentile interval under a halving (effect, $s = 0.5$) and the null ($s = 1$). Each entry uses 200 finite draws from one seed stream; nominal null rejection is 5%. Diagnostic runs are exploratory. Five-cell intervals are evaluated before the reporting gate, which abstains below 20 cells.

cohort / pool	5 cells		50 cells		100 cells		800 cells	
	effect	null	effect	null	effect	null	effect	null
SMC / pooled	70.5	63.0	52.5	28.5	56.0	18.0	69.5	9.5
SMC / reference	58.0	37.5	78.0	12.0	88.5	7.0	98.5	7.0
KUL3 / pooled	75.5	67.0	54.0	30.0	33.5	19.0	53.5	4.5
KUL3 / reference	70.0	61.5	55.0	15.5	56.0	16.0	87.0	6.5

At five cells, null rejection is 37.5–67.0%, far above the nominal 5%. For the pooled SMC and KUL3 alternatives, respectively 96 and 85 of 200 tumour samples have all-zero target expression; all of these exclude zero. Resampling such a tumour sample cannot recover its unobserved positive tail. The reference arm still varies, so none of the 6,400 full intervals has exactly zero width. The pooled five-cell median widths are 1.45 and 1.72, compared with 2.35 and 2.01 at 50 cells. These checks support sparse-sample undercoverage, rather than a claim that the whole interval collapses to a point. Even at 50 cells, null rejection is 12.0–30.0%. The original 90% benchmark-inclusion criterion at $s = 0.5$ therefore cannot substitute for checking the null.

Nonmonotonicity also changes candidate interpretation. In the full saved union grid, three of eight SMC/reference seeds fail a criterion at a larger count after first qualifying. Requiring every subsequently evaluated count to qualify changes first candidates 45, 50 and 50 to 60, 70 and 70. No such reversal occurs in the eight KUL3/reference seeds; neither pooled condition has a qualifying candidate. This is a descriptive re-reading of the same simulations, not an independently validated replacement selection rule. Together, the null diagnostic and reversals make the candidates useful as sensitivity summaries, not operational cutpoints. The diagnostic uses one seed stream and fixed observed cohorts; its binomial Monte Carlo standard errors are at most 3.54 percentage points, excluding patient uncertainty.

Simulation precision is not patient-level uncertainty. Enumerating all 45 and 15 two-patient holdout pairs separates patient omission from repeated simulation. Keying streams to the sorted pair identifiers keeps every retained pair fixed. Raising the inner bootstrap budget from 200 to 1,000 and 5,000 leaves the first cohort’s candidate at 70, but moves the second’s 400 to 300 and back to 400. Omitting one source patient changes the joint coverage-and-discrimination candidate for none of ten patients in the first cohort and three of six in the second (KUL01, KUL30 and KUL31). For the coverage-only candidate, the corresponding counts are five of ten and zero of six. These are cohort-conditional influence checks, not intervals for a larger patient population. We therefore report candidates, not an endorsed calibrated cutpoint. Appendix B records further biological comparability limits.

Table 3: Reference-only sensitivity with every two-patient holdout pair represented once per count and effect. The three inner budgets reuse identical outer samples. Omission counts compare with the baseline at 200 inner draws; all pairs involving the omitted patient are removed and retained pairs are unchanged. These are sensitivity summaries within the observed cohorts.

cohort	pairs	joint candidate, inner draws			omissions changing candidate	
		200	1,000	5,000	joint	coverage only
SMC	45	70	70	70	0/10	5/10
KUL3	15	400	300	400	3/6	0/6

The sensitivity windows cover counts 40, 50, 60, 70, 80, 90, 100, 110 in SMC and 200, 300, 400, 600, 800 in KUL3. These per-count summaries use a fixed pair schedule, so they are a separate experiment from the eight-seed union grid. Omitting KUL01 or KUL30 moves the joint candidate from 400 to 300; omitting KUL31 moves it to the lowest tested count, 200, with its lower boundary unobserved. The coverage-only baselines likewise occur at the lowest tested counts, 40 and 200. Their stability cannot locate a lower crossing outside those windows. Enumerating pairs improves transparency about source-patient influence, but overlapping pairs do not create additional independent patients.

Biological controls fail to establish mechanism. The pre-committed panel included compositional, intrinsic and retained-gene controls. In the primary cohort the retained-gene control changed by -1.000 to -0.951 relative to baseline, almost matching the compositional controls' -1.000 to -0.952 . The intrinsic control had only 0.04 CP10K, compared with 19.4 and 24.9 for the abundant compositional controls (counts per 10,000), making it uninformative for the intended test. A second cohort reproduced the pattern. Thus the application has two distinct unresolved problems: its interval rule lacks the recorded calibration support, and its control panel does not distinguish the proposed biological mechanisms. Improving one would not establish the other.

3 Explaining the recovery curve and specifying its scope

The earlier internal analysis compared estimated intrinsic effects with the requested reference in Equation 2. Ratios were 1.00–1.07 where the rule returned an estimate on reference-only draws and 0.86–1.18 on pooled draws, with shrinking dispersion at larger counts (Appendix B). The oracle estimator applies Equation 1 to the draw's own realised summaries. Its residual against that empirical decomposition is zero across 1,711,200 diagnostic replicates on two cohorts. The raw calculation also includes zero-cell cases; those arithmetic outputs do not define an intrinsic effect in an absent population (Appendix B). On defined contrasts, the recovery ratio therefore describes realised over requested reference. It does not measure whether the reporting gate meets its interval requirements.

The general form. Let D be a simulated draw requested at parameter θ . Pre-specify an observed-data reference $T(D) = h(S(D))$, where S denotes summaries of the draw. We call $T(D)$ the *realised reference*; it need not be a realised causal effect. If the estimate uses the same functional, $\hat{\theta}(D) = h(S(D))$, then, for $\theta \neq 0$,

$$\frac{\hat{\theta}(D)}{\theta} = \frac{T(D)}{\theta}, \quad r(D) = \hat{\theta}(D) - T(D) \equiv 0.$$

```

r = estimate[valid] - reference[valid]
max_residual = np.max(np.abs(r))

```

Here `valid` marks paired finite outputs; report excluded draws and do not compute the maximum if none remain. This is a numerical discrepancy check, not a validity screen. Report the raw maximum and outcome units; if using a closeness label, state both absolute and relative tolerances and any sensitivity to that choice. No universal tolerance separates correct from incorrect estimators. Only an algebraic argument establishes an identity. Sharing summaries alone is insufficient: different functions of the same summaries need not agree.

181 **What the audit can establish.** The recovery curve still measures sampling performance against θ . If
182 estimate and reference are equivalent, their curves are redundant for comparing those implementations.
183 That is compatible with an accurate estimator and a useful recovery experiment. The unsupported
184 inference is from this point-estimate comparison to interval coverage or an abstention rule, neither
185 of which the comparison evaluates. A zero residual therefore prompts inspection of the claimed
186 evidence; it is not a rejection criterion for an estimator.

187 **A control separating equality from performance.** To test this distinction directly, a separate
188 simulation fixes the reference before evaluating any estimator: for 200 independent observations
189 $Y_i \sim \mathcal{N}(\theta, 1)$, set $T(D) = \bar{Y}$. We evaluate four procedures at $\theta = 0$ and 0.5, with 2,000 replicates
190 per effect. The baseline reports the full-sample mean with a normal 95% interval using the sample
191 standard deviation. A second procedure keeps that point estimate but quarters the interval half-width.
192 A third adds a known bias of 0.5 to the baseline point and interval. The fourth uses every other
193 observation, selected before seeing outcomes, with its own mean and standard error. This half-sample
194 estimate shares observations with the full-sample reference; it is not independent of $T(D)$.

Table 4: Equality and statistical performance in the clean control. All 2,000 outputs per row are finite. Residuals compare points with the pre-specified full-sample mean; bias, RMSE and coverage compare with the requested θ . The nominal coverage is 95%.

θ	procedure	$\max r $	bias	RMSE	coverage
0	Full-sample mean	0	-0.0008	0.0710	94.60%
	Same point, narrow interval	0	-0.0008	0.0710	35.95%
	Mean plus 0.5	0.5000	0.4992	0.5042	0.00%
	Half-sample mean	0.2883	0.0004	0.0994	95.00%
0.5	Full-sample mean	0	0.0001	0.0710	95.40%
	Same point, narrow interval	0	0.0001	0.0710	36.30%
	Mean plus 0.5	0.5000	0.5001	0.5052	0.00%
	Half-sample mean	0.2686	-0.0017	0.1015	94.20%

195 Table 4 puts adequate and inadequate intervals in the same zero-residual class. Conversely, departure
196 contains both the unbiased half-sample mean and the deliberately biased estimate. Coverage Monte
197 Carlo standard errors are at most 1.08 percentage points. Neither residual class certifies performance,
198 even with a reference declared in advance. At the null, differences remain well-defined although
199 recovery ratios do not.

200 **Report performance and equivalence separately.** State the scientific target and whether it is fixed
201 or draw-dependent, then report target error and interval performance. If a realised reference is also
202 used, declare its functional and observation process before inspecting residuals. Report attempted and
203 valid pairs, maximum absolute residual in outcome units, and both tolerances used for a numerical
204 classification. Classify the result as a proved identity on a stated domain, numerical agreement on the
205 tested draws, or an observed departure; only the first supports an identity claim. Establish identities
206 algebraically on a stated class of datasets; agreement on a restricted, noiseless design need not extend
207 beyond it. Evaluate abstention frequency and conditional interval performance separately, using
208 independent evaluation draws for any selected rule. At null effects, report differences instead of
209 undefined recovery ratios.

210 4 Worked checks: what changes an empirical comparison?

211 The following examples use established relationships to check three choices: the reference, the
212 observation process and the fitted estimator. They are implementation evidence for interpreting a
213 recovery curve, not new equivalence theorems. Appendix A gives the simulation settings.

214 **Choose the reference before interpreting a residual.** For stratum counts n_g and arm means $\bar{Y}_{ag}$,
215 standardisation uses $T(D) = \sum_g (n_g/n)(\bar{Y}_{1g} - \bar{Y}_{0g})$. Saturated G-computation [Robins, 1986]
216 equals this functional by construction. Empirical-propensity IPW also equals it when both arms are
217 nonempty in each stratum: weighting the treated mean by $1/\hat{p}_g$, with $\hat{p}_g = n_{1g}/n_g$, gives $(n_g/n)\bar{Y}_{1g}$,

and similarly for controls. The augmentation terms of non-cross-fitted saturated AIPW sum to zero within stratum; matching with all controls recovers the same stratum means.

OLS with additive stratum indicators instead weights stratum contrasts by normalised $n_g \hat{p}_g(1 - \hat{p}_g)$ [Angrist, 1998], obtained by residualising treatment and outcome on stratum indicators [Frisch and Waugh, 1933]. Table 5 checks these identities numerically. Under confounding, G-computation and OLS switch between zero and 0.0663 residual when the reference changes. Complete 1:1 blocks make both weights proportional to n_g , whereas Bernoulli assignment balances only in expectation. The result is an instruction for interpreting the comparison: specify its weights and assignment constraints before calling two recovery curves independent evidence. The observed 10^{-14} residuals verify the implementation.

Table 5: Maximum absolute residual at 5,000 patients, with 1,200 replicates pooled over 6 seeds. Values near 10^{-14} reflect numerical equality. Changing the reference or using complete 1:1 blocks changes which estimators reproduce the reference. These are implementation checks of known identities and departures, not measures of estimator accuracy.

estimator	confounded		block-randomised	
	standardised	var-weighted	standardised	var-weighted
G-computation, generator’s strata	0	0.0663	0	9×10^{-16}
IPW, saturated propensity	1×10^{-14}	0.0663	7×10^{-15}	7×10^{-15}
AIPW, saturated, no cross-fitting	4×10^{-15}	0.0663	4×10^{-15}	4×10^{-15}
matching on stratum, all controls	9×10^{-16}	0.0663	9×10^{-16}	0
matching on stratum, 20 controls	0.052	0.067	0.037	0.037
matching on stratum, 1 control	0.164	0.182	0.170	0.170
OLS, stratum dummies	0.0663	3.5×10^{-14}	2×10^{-14}	2×10^{-14}
IPW, cross-fitted	0.599	0.618	0.182	0.182
unadjusted difference	4.99	4.99	7×10^{-15}	7×10^{-15}

Distinguish observed and latent quantities. With administrative censoring, the standardised exponential estimator uses events divided by exposure. Against that same observed-data functional its residual is exactly zero; against the corresponding latent-time functional, the maximum at 3,000 patients rises from zero without censoring to 0.376 at 78% censoring. The estimator has not changed; the comparison now includes information hidden by censoring. Conversely, standardised Kaplan–Meier RMST at $\tau = 2$, below every censoring horizon, agrees with the empirical latent truncated-mean contrast to 2.8×10^{-15} . Censoring outside the target horizon leaves this functional observable. Nonfinite fits are excluded in pairs; the most affected setting loses 463 of 1,200 draws (Appendix C). A larger latent residual does not rank estimator quality.

Separate generator complexity from estimator complexity. Saturated G-computation reproduces the drawn reference in all 30 learned-generator and 12 parametric-control settings because the equality holds for each dataset, regardless of its generator. A mixture M-step likewise has $\pi_k = n_k/n$ and $\mu_k = \sum_i r_{ik} y_i / n_k$, implying $\sum_k \pi_k \mu_k = \bar{y}$. These are explanatory identities, not tests of patient-model fidelity. Table 6 shows their scope: ridge and MLP outcome fits can depart, with an MLP maximum of 3.3 on mixture-generated data. The run contains 75,580 evaluations and 74,726 finite drawn-reference pairs. Of 420 settings, 41 ridge settings change numerical classification when relative tolerance is added. This demonstrates why the raw residual, outcome units and tolerances must accompany any closeness label (Appendix C).

External evidence bounds the claim. The pre-registered search found no functional reuse in seven classified public repositories and no examinable code for its virtual-control-arm query. This is neither a representative prevalence estimate nor corroboration of a widespread failure mode. The positive evidence in this paper is the documented internal case; the negative search result limits its external reach (Appendix D).

Table 6: Maximum absolute residual against the drawn reference, over cohort sizes and generators within each family, conditional on finite pairs. All ten estimators in the expanded run are shown. The saturated zero row is exact; other entries are rounded from the pinned primary table. Changing generator capacity preserves the saturated identities but can change other residuals.

Estimator	Mixtures	MLP generator	Parametric controls
G-computation, saturated	0	0	0
IPW, saturated	1.6×10^{-14}	1.2×10^{-14}	1.1×10^{-14}
G-comp., mixture outcome $K = 1$	9.3×10^{-15}	8.0×10^{-15}	8.0×10^{-15}
G-comp., mixture outcome $K = 4$	2.2×10^{-14}	1.6×10^{-14}	1.8×10^{-14}
G-comp., ridge $\alpha = 10^{-5}$	7.9×10^{-5}	6.3×10^{-5}	8.6×10^{-5}
G-comp., ridge $\alpha = 10^{-3}$	0.0079	0.00627	0.00864
G-comp., ridge $\alpha = 10^{-1}$	0.713	0.568	0.78
G-comp., MLP outcome	3.3	0.00908	0.0125
OLS, stratum dummies	1.51	1.42	1.71
Unadjusted difference	10.2	9.69	11.3

5 What a replacement validation would need to establish

The case identifies three decisions that must precede another cutpoint claim. First, fix the inferential target. A conditional cell-sampling study can hold a source pool fixed and evaluate intervals against $f_N(s-1)\mu_{\text{pool}}$. A patient-population study instead needs a patient-sampling model and a target averaged over that population. The current draw-dependent benchmark answers a third question. Results under one definition cannot be relabelled as coverage under another. The same distinction applies to a clinical simulator: requested population treatment effects, latent finite-cohort effects and empirical observed-data contrasts are separate validation targets.

Second, assess the whole decision rule using the interval that would actually be reported. Record bias and error against the chosen target, null rejection, detection at the specified effect, and abstention frequency. Report both conditional interval performance among returned answers and how often a valid answer is returned. Our five-cell diagnostic concerns arithmetic before a gate that already abstains there; the 50-cell null results concern its reporting boundary directly. A successful effect-detection rate can accompany excessive null rejection, so the former alone cannot validate an interval. A fixed 5% null target is appropriate here because the reported interval is nominally 95%; other operating targets must follow the intended use.

Third, separate selecting a rule from assessing it. If a scalar threshold is intended to guarantee performance above that threshold, qualification must address the evaluated upper range rather than only its first passing point. Reserve independent simulation draws for the final assessment, report rate uncertainty and the tested range, and distinguish this from transfer to new patients or cohorts. Increasing resamples from an empirical pool improves conditional numerical precision; it does not broaden that pool's biological support. For an external-control application, the additional question is comparability across data sources, which requires direct study-level validation such as the held-out comparisons of Ventz et al. [2019]. The present experiments motivate these requirements but do not perform that replacement validation.

6 Conclusion

The original recovery evidence did not establish the specified interval requirements for the 50-cell reporting rule. Direct evaluation reveals cohort and aggregation sensitivity and severe low-count null rejection, while the biological controls fail to separate the intended mechanisms. These findings concern a documented, limited application and its draw-dependent benchmark. They identify why its validation claim should be withdrawn, rather than a replacement clinical threshold.

The estimator identities explain what the original curve measures; they add no new statistical guarantee. The practical consequence is to specify the target and sampling unit, evaluate null rejection and effect detection with the same interval, and report abstention and conditional performance. Any revised rule then needs evaluation on independent draws and, for patient-level claims, independent patients.

287 7 Responsible use, limitations and impact

288 **Misuse and limitations.** NOT ESTIMABLE must not be read as “no effect” or as evidence that
289 surviving cells are behaving normally. A separate within-cell analysis over three cohorts and 42
290 patients also remained unresolved because its comparability checks returned *undecided* (Appendix B).
291 Calibration here covers one gene, one labelling axis and one weighting on two small cohorts.
292 It challenges this rule’s justification, not cell-count rules in general. Its stored coverage reference
293 depends on the sampled baseline, as specified in Equation 2. The draw-pool choice was underspecified,
294 repeated simulations do not add independent patients, and the leave-one-patient-out sensitivity
295 remains conditional on these cohorts; patient-level population uncertainty in the candidate cutpoints is
296 unquantified. The absence-defined labelling axis remains sensitive to sequencing depth (Appendix B).
297 The worked patient simulations are implementation checks. The external search found no affirmative
298 case and supplies no estimate of prevalence.

299 **Data and impact.** The single-cell cohorts are public, de-identified and previously published. The
300 release policy covers derived summaries and code, rather than cell-level matrices. A system reporting
301 group-specific quantities should distinguish an unavailable comparison from an estimated null and
302 should evaluate the coverage and discrimination of any abstention rule. Publishing failed calibration
303 attempts helps prevent unsupported thresholds from being treated as validated.

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389 A Simulation settings

Experiment	Fixed settings and sampling units
Single-cell calibration	GSE132465 (SMC): 10 paired patients; GSE144735 (KUL3): 6. Target: GUCA2A; <code>stem_pole</code> axis, <code>lineage</code> resolution, <code>differentiated</code> mature label; reference (<code>normal</code>) weighting. Each replicate holds out two patients and draws 2,000 cells per synthetic arm from their eligible pool; reference mature fraction 0.40. Expression shifts are 1.0, 0.8, 0.5, 0.25; calibration uses 0.5.
Interval construction	A nominal 95% percentile interval from 200 resamples of mature cells within each synthetic arm, holding cell fractions fixed. This assesses the conditional intrinsic contrast; it does not estimate population uncertainty across patients. Coverage and discrimination condition on defined intervals. Coverage uses $f_N(s-1)\bar{Y}_N$ with the drawn reference-arm mean, not a fixed pool mean.
Patient outcomes	Stratum probabilities (0.5, 0.3, 0.2), baseline means (10, 14, 25) and $Y \mid (g, A) \sim \mathcal{N}(\mu_g + 3A, 4^2)$. Confounded treatment probabilities (0.25, 0.50, 0.75); alternatives are independent Bernoulli assignment at 0.5 and complete balanced blocks of size 4 within strata. The three-design sweep uses 200 replicates per size and seed across six seeds.
Survival outcomes	The same stratum and confounded-assignment probabilities; exponential event rate $\lambda_g \exp(\beta A)$, $\lambda = (0.10, 0.16, 0.30)$, $\exp(\beta) = 0.6$. Administrative horizons 1000, 20.065216, 6.072920, 2.028858 target 0%, 12%, 49%, 78% censoring. The same latent cohort is viewed through each horizon; RMST uses $\tau = 2$. Each size has 200 replicates per seed over six seeds.

391 Settings are recorded by `calibration_gap`, `trial_recovery` and `trial_survival`; the dense
392 calibration grid is specified by `cutpoint_dense_grid`. Historical grid budgets are reported in
393 Appendix B; Section 2 specifies the controlled runs. These are existing simulations, not clinical-trial
394 data.

395 B Calibration details and limits

396 **Historical grids.** Before the controlled run, the SMC reference-only sweeps returned binned
397 joint candidates of 100, 90 and 70 on the committed, filled and dense grids, with 500, 500 and
398 200 replicates per setting, respectively. Their adjacent crossing brackets were (40, 100], (65, 90]
399 and (42.5, 70], describing evaluated bin summaries rather than confidence intervals. The 100-to-90
400 change occurred within the same bin, (76, 107]: adding counts 80 and 90 to 100 changed both its
401 median count and its aggregated outcomes. Random streams and bin boundaries also differed, so this
402 comparison alone did not isolate a resolution effect. The controlled analysis in Section 2 supersedes
403 it for that question. Under pooled draws no joint cutpoint qualified in either cohort’s historical grids.
404 KUL3 reference-only candidates were 400 or 800; its largest setting sampled 800 mature cells per
405 arm with replacement from 844 distinct mature source cells. Sampling with replacement is a valid
406 empirical-pool simulation, but an 800-draw arm is not 800 distinct cells or additional independent
407 patients.

408 **Recovery accounting and unavailable contrasts.** The four diagnostic sweeps comprise 1,711,200
409 replicates: 1,283,400 with nonzero requested intrinsic reference and 427,800 with zero requested
410 reference, for which ratios are undefined. The first 50-replicate sweep has 119,600 draws, partitioned
411 into 89,700 and 29,900, respectively. The maximum estimate–realised-reference residual is zero
412 throughout. At a zero-cell setting the raw ratio is $1/(1-s)$ when the denominator is nonzero. These
413 raw diagnostic values arise from the harness setting an empty-arm mean to zero; they do not define an
414 effect in absent cells. The reporting gate abstains and the interval is undefined. Separately, undefined
415 intervals were previously counted as failures through comparisons with NaN; corrected rates exclude
416 them and report abstentions. An all-abstaining bin has no coverage estimate. One historical seed
417 gives $ok = wide = 45$, showing that a three-state specification need not yield three distinct fitted
418 regions.

419 **Biological scope and patient uncertainty.** The finest annotation resolution yields abstention in
420 28/28 patients under one population definition and 24/28 under the other; coarser definitions do
421 not resolve identifiability. The coarsest definition labels every epithelial cell mature and forces the

compositional term to zero. In a separate comparability analysis over three cohorts and 42 patients, 83 of 84 housekeeping-detection control rows pass a 0.10 tolerance despite expression changes of 1.19–1.64 fold. Patient-bootstrap $\log_2$ expression intervals cross the accepted $[-0.5, 0.5]$ boundaries in every cohort, leaving the comparisons undecided. Depth matching discards roughly three quarters of the cells and pushes about half the remaining estimable rows below the count threshold. The two source pools also differ jointly in tissue mixture, dispersion and distinct mature-cell counts (662 versus 3,824); their comparison cannot isolate these factors. These observations restrict biological interpretation. Repeated cell draws and overlapping patient pairs do not supply independent patient replication.

C Supporting trial and generator checks

Reference dependence beyond the main comparison. For logistic-propensity IPW, confounded-design maximum residuals against the standardised reference are 0.403, 0.045, 0.0103 and 0.010 at $C = 0.1, 1, 100, 10^6$; tightening solver tolerance to 10^{-8} reduces the last to 1.3×10^{-6} . Under increasing effect heterogeneity ($\tau = 0$ to 16, fixed population ATE), OLS recovery grows from 1.003 to 1.132, while its residual against its own variance-weighted reference stays at numerical precision. Recovery can therefore reveal target mismatch even when functional equality persists. These results supplement Table 5.

Residual scale. The optional descriptor $\rho = \text{med}|\hat{\theta} - T| / \text{med}|T - \theta|$ compares typical departure with reference error when its denominator is positive. It is zero at exact equality and undefined when the denominator is zero. Because the two terms in $\hat{\theta} - \theta = (\hat{\theta} - T) + (T - \theta)$ can covary or cancel, ρ neither decomposes total error nor ranks estimators. A zero median can also conceal nonzero residuals, so it cannot replace the maximum-residual check.

Independent Cox implementation check. The survival sweep uses 1,200 paired draws per setting; at 78% censoring and 100 patients, 463 have nonfinite fits and are excluded. Eight stratified and unadjusted Cox fits on one fixed 1,500-patient cohort at four censoring horizons agree with `lifelines.CoxPHFitter` to 1.4×10^{-7} , below the specified 10^{-5} tolerance. All eight pairs are finite. This checks the implemented coefficients on that cohort, not coverage or convergence over the simulation grid.

Learned-generator design and exclusions. The pre-registration preceded the module; a recorded amendment added penalised outcome models after the original estimator set could not test the proposed near-equality interval. Seven generators (mixtures at $K = 1, 2, 4, 8$, an MLP conditional Gaussian and two parametric controls), ten estimators, six cohort sizes, six seed streams and 30 replicates schedule 75,600 evaluations on 7,560 synthetic cohorts (seed 20260913). After generator failures the saved table contains 75,580 evaluations on 7,558 cohorts. Against the drawn reference, 74,726 pairs are finite and 854 are excluded: 410 have nonfinite references and 731 nonfinite estimates, overlapping in 287 evaluations. All 420 generator-by-estimator-by-size cells retain valid pairs. Table 6 uses maxima within each generator family, conditional on those pairs.

Tolerance and the model-based target. NumPy’s default comparisons use 10^{-8} against a zero residual and $10^{-8} + 10^{-5}|T|$ for estimate versus reference. Thirteen generator–estimator combinations fail both at $n = 100$ but pass only the second at $n = 5,000$; all use ridge penalties. In the saturated ridge model, fitted means are $\bar{y}_{ga}n_{ga}/(n_{ga} + \alpha)$, so a fixed penalty diminishes with cell count. This numerical transition does not imply declining statistical validity. The fitted generator’s population effect T_{model} is a different reference from the empirical $T(\tilde{D})$. Mixture references use 200,000 Monte Carlo observations; saved median standard errors are 0.019–0.027. Parametric and MLP references are analytic. In $\hat{\theta} - \theta = (\hat{\theta} - T_{\text{model}}) + (T_{\text{model}} - \theta)$, estimator departure and generator target discrepancy can covary. The finite Monte Carlo budget limits numerical resolution, not the target’s intrinsic precision; neither reference validates a model for real patients.

D External audit and research context

Search and classification. A pre-registered queue contained 41 entries; 19 were screened out and seven classified within the 26 entries examined in order. An amendment before code inspection relaxed unsuccessful queries; the classification rule was unchanged. Generating-parameter references or quantities computed outside the estimator’s observed records give NO. Same-record references require examination of inputs, weights, conditioning, nuisance fitting and observation process; YES requires affirmative evidence of functional identity. All seven recorded verdicts are NO: `causal`, `target-trial-emulation`, `ecasim`, `PlasmodeSimulation`, `StatsPAI`, `diff-diff`, and `Auto-Empirical-Research-Skills`. The first six use generating parameters or outside-sample quantities; the last compares cross-fitted partialling-out with full-sample OLS. Its 4.0×10^{-14} residual on noiseless linear data becomes 0.125 after adding outcome noise. The functions agree on the restricted noiseless support but not on general noisy datasets; the NO verdict concerns shared functional form. Classification used one reader, only this last repository was executed, and the sample is nonrandom. The virtual-control-arm query located no examinable code. These limits preclude a prevalence estimate.

Related scope. Cell-count filters are established in pseudobulk analysis [Crowell et al., 2020]; `scCODA`, `Milo`, `propeller` and `DA-seq` address composition [Büttner et al., 2021, Dann et al., 2022, Phipson et al., 2022, Zhao et al., 2021], while `cocoa` separates composition and expression changes [Petukhov et al., 2022]. Selective prediction motivates evaluating both answers and abstentions [Chow, 1970, El-Yaniv and Wiener, 2010, Geifman and El-Yaniv, 2017]. Conformal prediction coverage [Vovk et al., 2005, Angelopoulos and Bates, 2023] does not define a missing conditional population or automatically provide intervals for this intrinsic parameter.

Documentary source of the case. The quoted validation claim and its six-draw recovery table occur in `docs/gate_memo_w2.md`, Section 15. The interval requirements occur in `docs/harness_design_spec.md`, Section 4. The former declares the 50-cell cutpoint validated; the latter requires 90% inclusion and 80% exclusion of zero at $s = 0.5$. Section 2 reproduces the relevant claim and numbers so this comparison can be inspected without access to an identified repository. These are internal documents, not a published clinical validation. Checksums and exact source excerpts accompany the revision.

Implementation and provenance. Corrections separated population definitions collapsed by deduplication, derived fractions and means from one labelling call, tracked producing scripts, and repaired undefined-interval denominators. Saved tables retain commit hashes and seeds. These implementation checks support traceability, not statistical validity; the draw-dependent calibration target is stated in Equation 2.

Use of large language models. Language-model assistants supported coding, figures, drafting and model audits under author direction. They helped identify defects and introduced some of those above. Authors remain responsible for all claims. Automated checks re-derive selected results and complement review; they do not verify every manuscript statement.